

STEP I - 1988

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Abstract

A collection of solutions to STEP questions for my own use and also for former colleagues and friends. These are very much incomplete. I aim to complete some sporadically although I make absolutely zero promise of doing so with any sort of regular schedule. I largely do this for fun. Although I try my best, I have undoubtedly made several errors so please do use with caution. Any errors, please do let me know at ccbarnesni@gmail.com.

1. Since $x > 0$ the graph must never cross the y -axis. We then calculate the turning point

$$h'(x) = \frac{1 - \ln x}{x^2}$$

We see that $h'(x) = 0$ when $x = e$ only and the corresponding y -coordinate is $1/e$. The graph must be negative for $x < 1$ since $\ln x < 0$ for $x < 1$ but $1/x > 0$ for $0 < x < 1$. It is therefore increasing before the turning point so we anticipate it to be a maximum. To verify, we check the 2nd derivative

$$h''(x) = \frac{2 \ln x - 3}{x^3}$$

and see that $h''(e) < 0$ so $(e, 1/e)$ is indeed a maximum. The graph must remain positive from this point onwards thus $x = 0$ is an asymptote.

To solve $n^m = m^n$ for $n, m \in \mathbb{N}$ we note, through rearrangement, that this is equivalent to solving

$$\frac{\ln n}{n} = \frac{\ln m}{m}.$$

We are thus looking for where on the graph $h(x)$ two y -coordinates can be equal. The only possible place where this can occur is when $1 < x < e$ since this is the only region in the graph where a given point on $h(x)$ has another corresponding point with equal y -coordinate. There is only one integer in this interval, namely, 2. We thus seek solutions to

$$\begin{aligned} \frac{\ln 2}{2} &= \frac{\ln x}{x} \\ \iff x \ln 2 &= 2 \ln x \\ \iff 2^x &= x^2 \end{aligned}$$

By considering the intersection of the graphs $y = x^2$ and $y = 2^x$ we see that there can only be one positive point of intersection. By inspection, we see that this value is 4. Thus the only pairs $m, n \in \mathbb{N}$ satisfying the required equation are 2 and 4.

2. Using the chain and/or quotient rule we obtain the following. We omit the independent variable for clarity.

$$\begin{aligned} g &= f + \frac{1}{f} \\ g' &= \frac{f'f^2 - f'}{f^2} \\ g'' &= \frac{f''f^4 - f''f^2 - 2f(f')^2}{f^4} \end{aligned}$$

We see that $g' = 0$ only when $f'f^2 - f' = 0$. Factorising, we see that

$$f'(f - 1)(f + 1) = 0.$$

We solve all three cases. Start with $f' = 0$. We have

$$\begin{aligned}\sin 2x - \cos x &= 0 \\ 2 \sin x \cos x - \cos x &= 0 \\ \cos x(2 \sin x - 1) &= 0.\end{aligned}$$

The only solutions to this in the interval $[0, 2\pi]$ are $\pi/2, 3\pi/2, \pi/6, 5\pi/6$.

We now consider when $f = 1$. We have

$$1 - 2 \sin^2 x + 2 \sin x + 4 = 1 \tag{1}$$

$$\sin^2 x - \sin x - 2 = 0 \tag{2}$$

$$(\sin x - 2)(\sin x + 1) = 0. \tag{3}$$

Clearly $\sin x = 2$ has no solutions and $\sin x = -1$ yields $x = 3\pi/2$ as the only solution.

Now we turn to $f = -1$. In this case, we have

$$\begin{aligned}4 + 1 - 2 \sin^2 x + 2 \sin x &= -1 \\ \sin^2 x - \sin x - 3 &= 0 \\ \sin x &= \frac{1 \pm \sqrt{13}}{2}.\end{aligned}$$

This also has no solutions since $|\frac{1 \pm \sqrt{13}}{2}| > 1$. So we have now found all 4 stationary points.

We verify if these are maxima or minima by considering the sign of g'' . To ease computation we first compute the values of f, f', f'' at each of the respective coordinates. Note that $f' = 0$ at all points of interest. We then find that

$$\begin{aligned}f(\pi/2) &= 5 \\ f''(\pi/2) &= 2 \\ f(3\pi/2) &= 1 \\ f''(3\pi/2) &= 6 \\ f(\pi/6) &= \frac{11}{2} \\ f''(\pi/6) &= -3 \\ f(5\pi/6) &= \frac{11}{2} \\ f''(5\pi/6) &= -3\end{aligned}$$

It is then easy to compute that $g''(\pi/2) = \frac{2(5^4) - 3(5^2)}{5^4} > 0$ implying a minimum. We also have

$$g''(3\pi/2) = 0$$

which we note, and return to later. Then

$$g''(\pi/6) = g''(5\pi/6) = \frac{3(11/2)^2 - 3(11/2)^4}{(11/2)^4} > 0$$

again giving us two minimums.

We now return to the case of $x = \pi/6$. To examine this, we look at the first derivative and find that the sign of g' depends on the sign of f' and $f^2 - 1$ only. For $\epsilon > 0$ sufficiently small, we see that

$$f'(\pi/6 \pm \epsilon) = 2 \cos\left(\frac{\pi}{6} \pm \epsilon\right) \left(1 - \sin\left(\frac{\pi}{6} \pm \epsilon\right)\right) > 0.$$

Then,

$$f^2\left(\frac{\pi}{6}\right) - 1 = \left(\frac{11}{2}\right)^2 - 1 > 0$$

and so $f^2 - 1$ remains positive in some neighbourhood around $\pi/6$. Thus g' remains positive in this region too and so we have a point of inflection.

We have 3 minimia and 1 point of inflection - something must definitely have gone wrong here. Actually, it could be due to there being asymptotes when $f = 0$.

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9. Using integration by parts we see that

$$\begin{aligned} & \int_1^e \frac{\ln x}{x} dx \\ &= \left[-\frac{\ln x}{x} \right]_1^e - \int_1^e -\frac{1}{x^2} dx \\ &= -\frac{1}{e} + \left[-\frac{1}{x} \right]_1^e \\ &= -\frac{1}{e} - \frac{1}{e} + 1 \\ &= 1 - \frac{2}{e} \end{aligned}$$

For the second integral, consider the substitution $u = \sqrt{1 + \sin x}$. Then $du = \frac{\cos x}{2\sqrt{1 + \sin x}} dx$

and $u^2 - 1 = \sin x$. Then we have

$$\begin{aligned}
 & \int \frac{2}{u^2 - 1} du \\
 &= \int \frac{1}{u - 1} - \frac{1}{u + 1} du \\
 &= \ln |u - 1| - \ln |u + 1| + c \\
 &= \ln \left| \frac{u - 1}{u + 1} \right| + c \\
 &= \ln \left| \frac{u^2 - 2u + 1}{u^2 - 1} \right| + c \\
 &= \ln \left| \frac{\sin x + 2 - 2\sqrt{1 + \sin x}}{\sin x} \right| + c
 \end{aligned}$$

10. If the horizontal distance covered is d and the angle of projection is θ then we must have

$$d = vt \cos \theta.$$

Vertically, if we neglect air resistance, we have

$$h = ut \sin \theta - \frac{g}{2} t^2$$

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